Prime Divisors of Fermat Numbers: Congruences and Infinitude

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Abstract

Let $F_n = 2^{2^n} + 1$ for $n \geq 0$. We give a self-contained derivation of the classical pairwise-coprimality identity and Lucas's congruence: every prime $p \mid F_n$, with $n \geq 2$, satisfies $p \equiv 1 \pmod{2^{n+2}}$. It follows that for every $m \geq 1$ there are infinitely many distinct primes congruent to 1 modulo 2^m. These facts concern divisors of Fermat numbers, not the unresolved question of which Fermat numbers themselves are prime.

Definitions and statement

OEIS A000215 comprises the Fermat numbers $F_n = 2^{2^n} + 1$ indexed from $n = 0$ [1]. Here $\mathrm{ord}_p(2)$ denotes the multiplicative order of 2 modulo an odd prime p.

Theorem 1. *For $n \geq 1$,*

$$\prod_{j=0}^{n-1} F_j = F_n - 2. \tag{1}$$

The numbers $F_0, F_1, \ldots$ are pairwise coprime. Furthermore, for every $n \geq 2$ and each prime divisor p of F_n,

$$p \equiv 1 \pmod{2^{n+2}}. \tag{2}$$

Proof. Repeatedly factor a difference of squares:

$$\prod_{j=0}^{n-1} (2^{2^j} + 1) = \frac{2^{2^n} - 1}{2 - 1} = F_n - 2.$$

If $0 \leq i < n$, then $F_i \mid F_n - 2$ by (1); hence $\gcd(F_i, F_n) \mid 2$. Every F_i is odd, so this gcd is 1.

Now fix $n \geq 2$ and an odd prime $p \mid F_n$. We have

$$2^{2^n} \equiv -1 \pmod{p}, \qquad 2^{2^{n+1}} \equiv 1 \pmod{p}.$$

As the order of 2 divides 2^{n+1} but not 2^n, it must be exactly

$$\mathrm{ord}_p(2) = 2^{n+1}. \tag{3}$$

By Fermat's little theorem, $2^{n+1} \mid (p - 1)$; because $n \geq 2$, this yields $p \equiv 1 \pmod 8$. The classical supplementary law for quadratic residues says $(2/p) = 1$ for $p \equiv \pm 1 \pmod 8$ [2]. Consequently Euler's criterion gives

$$2^{(p-1)/2} \equiv 1 \pmod{p}.$$

By (3), $2^{n+1} \mid (p - 1)/2$, whence $2^{n+2} \mid (p - 1)$, as claimed. □

Corollary 1. *For each fixed integer $m \geq 1$, infinitely many distinct primes p satisfy $p \equiv 1 \pmod{2^m}$.*

Proof. For every $n \geq \max\{2, m - 2\}$ choose a prime $p_n \mid F_n$. By Theorem 1, $p_n \equiv 1 \pmod{2^{n+2}}$, hence modulo 2^m. Coprimality of different F_n makes all p_n distinct. □

Scope, boundary cases, and priority

The restriction $n \geq 2$ in (2) is essential: $F_0 = 3$ and $F_1 = 5$ do *not* satisfy the displayed congruence with their respective indices. For example, $F_5 = 641 \cdot 6700417$, with each factor $\equiv 1 \pmod{128}$. The congruence is Lucas's classical theorem, explicitly recorded in A000215 and A023394 and discussed in the literature [1, 2, 3]. No novelty or proof of any Fermat-prime conjecture is claimed. In particular, whether there are Fermat primes beyond F_4, or only finitely many, remains unsettled [2].

Verification disclosure

This document was drafted and mathematically checked by an AI system; no human peer review is represented. Finite factorization checks for $0 \leq n \leq 6$ are consistency tests only, not substitutes for the argument above.

References

[1] OEIS Foundation, *A000215: Fermat numbers*, `https://oeis.org/A000215`; see also *A023394: Prime factors of Fermat numbers*, `https://oeis.org/A023394` (accessed November 12, 2026).

[2] K. Conrad, *Fermat Numbers and Their Factors*, `https://kconrad.math.uconn.edu/blurbs/ugradnumthy/fermat-numbers-and-factors.pdf`, Theorem 2.2 (Lucas).

[3] J. Lahtonen and M. Lavrov, answers to *How to prove that if a prime p divides a Fermat number then* $p = k2^{n+2} + 1$*?*, Mathematics Stack Exchange, `https://math.stackexchange.com/questions/91413`.